UNDER REVIEW

RESCIENCE C

Replication / Computational Neuroscience

[Re] Dreaming neural networks: forgetting spurious memories and reinforcing pure ones

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Edited by
(Editor)

Received
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Published
–

DOI
–

1 Introduction

Fachechi, Agliari and Barra[1] (Neural Networks 112, 2019, 24–40; arXiv:1810.12217) revisit the "unlearning" idea of Hopfield, Feinstein and Palmer[2] and derive a closed-form "dreaming" kernel for the Hopfield model. Starting from the Hebbian coupling matrix $J = \frac{1}{N}\sum_{\mu=1}^{P} \xi^{\mu}\xi^{\mu\intercal}$ built from $P = \alpha N$ random ± 1 patterns, a sleep phase of duration t produces

$$J(t) = J\,(1+t)\,(\mathbb{1} + tJ)^{-1},$$

which equals J at $t = 0$ and tends to the projection (pseudo-inverse) rule of Kanter and Sompolinsky[3] as $t \to \infty$. The article's statistical-mechanics analysis predicts that the critical storage load $\alpha_c(t)$ rises from the Amit–Gutfreund–Sompolinsky value[4] ≈ 0.138 at $t = 0$ towards the theoretical bound $\alpha = 1$, that spurious (mixture) attractors are removed, that pure patterns are reinforced, and that the gain is robust to thermal noise. The original Monte Carlo checks (Section 2.5 of the published article) use, for the magnetisation curves, single-spin-flip Glauber dynamics measured after equilibration on networks of $N = 5\,000$ neurons averaged over $M = 10$ pattern realisations (their Fig. 7, loads $P/N \in \{0.02, 0.08, 0.32\}$, $t \in \{0, 0.1, 1, 10, 100, 1000\}$) and a finite-size comparison at $N \in \{500, 1\,000, 5\,000\}$ (their Fig. 8). The zero-noise internal-field and basin analyses (their Figs. 9–10) use the *parallel* sign update of their Eq. (2.17) at $N = 1\,000$, $P = 50$. For the fields (Fig. 9) the network starts from random configurations, 20 evolutions for each of 20 pattern realisations; for the basins (Fig. 10) it starts from the first stored pattern with each bit flipped independently with probability p, success being declared when that pattern carries the largest Mattis overlap, 200 evolutions for each of 200 realisations. The replica-symmetric capacity rises from $\alpha_c(0) \simeq 0.138$ to $\alpha_c \simeq 1.07$ as $t \to \infty$, the excess over Gardner's bound being attributed to replica-symmetry breaking. No capacity curve is measured by Monte Carlo in the original: the simulations verify magnetisations, fields and basins against the theory.

We independently implement the published kernel and *partially* replicate its qualitative claims at finite size ($N \leq 1\,000$) with a new, noisy-cue capacity estimator and new random seeds: we test the retrieval gain, the removal of spurious final states, the robustness to thermal noise and the projector limit, and we do not reproduce the replica-symmetric capacity curve nor the original Monte Carlo figures. Each of the three studies was pre-registered and frozen by an annotated git tag before its own main run; the later registrations were informed by the preceding studies (Section "Pre-registration").

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The authors have declared that no competing interests exist.
Code is available at https://github.com/hypneum-lab/rescience-nh01..
Data is available at https://github.com/hypneum-lab/rescience-nh01/tree/main/data – DOI 10.5281/zenodo.23833431.

The replication was carried out inside the Hypneum Lab research programme (issue hypneum-lab/nerve-wml#68), where the dreaming kernel serves as a reference operation; the replication stands on its own and makes no claim about that programme.
Beyond replicating, we add one control absent from the original work: a uniform multiplicative rescaling ("downscale") of the Hebbian couplings, matched to the dreamed kernel on the Frobenius norm. This control answers a question the original article does not ask — whether the benefit of dreaming could be reproduced by a global reduction of synaptic strength, as in homeostatic-downscaling accounts of sleep[5] — and it provides a calibrated null for the capacity ratio. Finally, we test the kernel on correlated patterns, which the original article treats only for independent patterns.

2 Methods

2.1 Original model

Binary units $s_i \in \{\pm 1\}$, $i = 1 \dots N$; patterns $\xi^\mu \in \{\pm 1\}^N$ i.i.d. uniform; Hebbian kernel as above; dreamed kernel $J(t)$. In pattern space, $J_{ij}(t) = \frac{1}{N} \sum_{\mu\nu} \xi_i^\mu [(1+t)(\mathbb{1} + tC)^{-1}]_{\mu\nu} \xi_j^\nu$ with $C_{\mu\nu} = \frac{1}{N} \xi^\mu \cdot \xi^\nu$. The eigenvalues $\lambda \geq 0$ of J map to $\lambda(1+t)/(1+t\lambda)$. Retrieval is assessed by the overlap $m = \frac{1}{N} \sum_i s_i \xi_i^\mu$ of the final state with a stored pattern.

2.2 Implementation

Our code (S1: `run.py`, `confirmatory.py`, `correlated.py`; 713 + 384 + 237 lines, NumPy only, no original code reused) computes $J(t)$ by spectral decomposition of the symmetric J (`numpy.linalg.eigh`, negative numerical eigenvalues clipped to 0). Twenty test functions (23 executed cases) check that $J(0) = J$, that the spectral form equals the pattern-space closed form on 2-pattern and many-pattern cases, that $t = 10^8$ approximates the projector within the tested tolerance, that the matched shrink is exactly inert at zero temperature (S2), and cover the confirmatory and correlated runners. The diagonal J_{ii} is set to 0 in every arm before the dynamics. The dynamics are asynchronous with one fresh random permutation of the N units per sweep, shared by the probes of a cell. At $T = 0$, $s_i \leftarrow \mathrm{sign}(h_i)$ with ties ($|h_i| \leq 10^{-12} N \max|J|$) resolved by keeping the current spin — a tolerance chosen so that the rule is exactly invariant under $J \to cJ$; the run stops at the first sweep without a flip, cap 100 sweeps (non-convergence counted as not retrieved; none occurred in any run). At $T = 0.2$, Glauber updates with $P(s_i = +1) = 1/(1 + e^{-2\beta h_i})$, $\beta = 5$, 40 sweeps, overlap averaged over the last 10.
Cue: a stored pattern with exactly round$(0.10\,N)$ bits flipped (overlap 0.80 at $N = 1\,000$; 0.8125, 0.797, 0.797 at $N = 64$, 128, 256 because of rounding). Probes: the first $\min(P, 32)$ stored patterns, i.e. 10 and 20 probes at the loads 0.01 and 0.02 of the correlated run. The pilot also includes a descriptive $T = 0$ condition with $f = 0$ (unperturbed stored-pattern cues), absent from the two $N = 1\,000$ studies. Final states are classified as `retrieved` ($m \geq 0.9$; 0.8 at $T = 0.2$), `other_stored` or `spurious`. Capacity estimator per (arm, seed): $\hat{\alpha}_c$ is the linearly interpolated 0.5 crossing of the retrieval fraction $r(\alpha)$ immediately after the *last* grid load with $r \geq 0.5$, on the grid $\alpha \in \{0.04, 0.06, \dots, 0.20, 0.25, \dots, 0.95\}$ (24 values; 0.01 and 0.02 added for the correlated run). If the largest load still has $r \geq 0.5$ the estimate is right-censored at 0.95; if no load reaches 0.5 the code records the flag `below_grid` and a zero sentinel, which marks censoring, not a measured zero capacity. Patterns are nested in α and the same patterns, cues and update orders are used for every arm of a cell (paired design). Statistics are paired per-seed ratios and differences, mean over 10 seeds with a 95 % percentile bootstrap CI (10 000 seed-level resamples, bootstrap seed 20261025). Every random draw comes from `numpy.random.default_rng([20261025, seed, N, stream])`.

2.3 Matched uniform-shrink control (added)

For each $t > 0$, $J_c = c(t)\,J$ with $c(t) = \|\text{offdiag}\,J(t)\|_F / \|\text{offdiag}\,J\|_F$, i.e. equal global synaptic budget; c depends on t, α, N and the seed and the correlation level; across the shrink arms it ranges from 0.0884 to 0.9985. Two facts were declared before any run so that the control cannot be misread: at $T = 0$ the sign dynamics is invariant under $c > 0$, so the control must give a ratio of exactly 1 (an implementation check, not an empirical hypothesis); at $T > 0$, J_c at inverse temperature β has the same transition probabilities as J at $c\beta$, i.e. the control is the Hebbian network at temperature T/c. This identity does not impose a deterministic inequality on a finite-time, thresholded, single-realisation capacity estimate (in the pilot, seed 6 at $N = 256$, $T = 0.2$ gives a shrink ratio of 1.012), so the effect at $T = 0.2$ is measured empirically.

2.4 Correlated patterns (added)

One ancestor $a \in \{\pm 1\}^N$ per (seed, level); each pattern bit equals a_i with probability $(1 + m)/2$, so the mean pairwise overlap is m^2. Levels $m \in \{0, 0.15, 0.30\}$ ($q \in \{0, 0.0225, 0.09\}$), seeds 100–109 (disjoint from the independent-pattern seeds 0–9).

2.5 Pre-registration and decision rules

Three L1 pre-registrations, each frozen by an annotated tag pushed to GitHub before launch and each archived by Software Heritage (verbatim copies in `prereg/` of the submission repository, S3):

- **Pilot** (tag of 2026-10-25, S3): $N \in \{64, 128, 256\}$, $t \in \{0, 0.3, 1, 3, 10, 30, 100\}$, three conditions.
- **Confirmatory** (second tag of 2026-10-25): $N = 1\,000$, $t \in \{0, 1, 3, 5, 7, 10, 15, 20, 30, 100\}$, $T \in \{0, 0.2\}$.
- **Correlated** (tag of 2026-10-27): as confirmatory, three correlation levels, 26 loads.

Binding criteria at $t = 10$: **C1** (gain) confirmed if the CI lower bound of the ratio R of the dreamed ($t = 10$) to the Hebbian capacity at $T = 0$ is ≥ 3, refuted if the CI upper bound of the paired difference is < 0.1; **C2′** (shape) holds if the median curve $t \mapsto \hat{\alpha}_c$ is unimodal within 0.02 with an interior maximum t^*; **C3′** (control) holds if the CI upper bound of the ratio S of the shrink ($c(10)$) to the Hebbian capacity at $T = 0.2$ is < 1.1, refutes if the lower bound is ≥ 2; the $T = 0$ ratio must be exactly 1 on every seed, otherwise the run is void. **P4** (secondary label): spurious final states at $\alpha \in \{0.14, 0.20\}$ reduced by $\geq 50\,\%$. **C4** (correlated run): paired difference $R(m = 0.30) - R(m = 0)$ with CI excluding 0. Seeds whose Hebbian $\hat{\alpha}_c$ is below the grid have an undefined ratio.
These are the confirmatory and correlated criteria. The pilot registration used the original C2 (monotonicity of the median curve, tolerated decrease 0.05 per grid step), C3 at $T = 0$ only, and P4 at loads 0.10 and 0.14; the confirmatory amendment replaced C2 by C2′ and moved C3 to $T = 0.2$ after the pilot had been observed, and the correlated registration was written after both independent-pattern studies. The pilot's median curve decreases from 0.600 to 0.522 between $t = 10$ and $t = 100$ and passed C2 only because no single step exceeded 0.05.

2.6 What was changed with respect to the original, and why

1. Finite $N \leq 1\,000$ and an empirical, noisy-cue, 0.5-crossing capacity estimator instead of the replica-theory $\alpha_c(t)$: absolute capacities are therefore not comparable with the published curve; the replication tests ratios and qualitative claims (gain, spurious removal, projector limit).

2. Self-coupling removed ($J_{ii} = 0$) in all arms: standard for retrieval dynamics; it changes the update fields relative to the original rule, whose field sums over $j = 1 \dots N$. We did not run an ablation with the diagonal kept, so its contribution to the finite optimum in t reported below is not isolated.
3. Dynamics and success criterion: asynchronous updates at $T = 0$ instead of the parallel sign update of the original zero-noise field and basin experiments (their Eq. 2.17); a fixed number of flipped cue bits, round($0.10\,N$), instead of independent flips with probability p; an absolute overlap threshold (0.9, or 0.8 at $T = 0.2$) instead of "largest Mattis overlap"; at $T = 0.2$, a fixed 40-sweep window without an equilibration diagnostic.
4. Matched shrink control and correlated patterns: additions, not changes.
5. Parameters and decision rules fixed before each study's main run (grids, seeds, thresholds); one voided launch of the pilot (BLAS thread oversubscription, killed after ~150 s before the main run; log kept in the upstream repository, D1).

3 Results

3.1 Independent patterns

Pilot (D1), $N = 256$, $T = 0$, $t = 10$: Hebb $\hat{\alpha}_c = 0.175$ [0.167, 0.183] → dream 0.599 [0.591, 0.606], ratio **3.43 [3.26, 3.60]**, difference 0.423 [0.411, 0.435]; shrink ratio 1.000 on every seed, every N, every t. Ratios at $N = 64$ and 128: 3.13 [2.74, 3.57] and 3.38 [3.21, 3.53]. Confirmatory (D2), $N = 1\,000$, 10 seeds, 9 120 cells:

Criterion (cell)	Result	Label
C1 gain ($T = 0$, $t = 10$)	Hebb 0.157 [0.155, 0.160] → dream 0.607 [0.604, 0.610]; ratio **3.86 [3.79, 3.93]**; difference 0.450 [0.445, 0.454]	confirmed
C2′ shape ($T = 0$)	medians 0.158, 0.413, 0.571, 0.603, **0.616**, 0.606, 0.595, 0.582, 0.569, 0.527 for t = 0 … 100; $t^* = 7$ (9 seeds at 7, 1 at 10); drop $\hat{\alpha}_c(7) - \hat{\alpha}_c(100)$ = 0.088 [0.082, 0.092]	holds
C3′ control ($T = 0.2$, $t = 10$)	Hebb 0.126 [0.125, 0.128] → shrink 0.123 [0.121, 0.125]; ratio **0.975 [0.967, 0.982]**; $T = 0$ ratio exactly 1 on every seed and t	holds
P4 spurious ($T = 0$)	$\alpha = 0.14$: 0.159 → 0.000; $\alpha = 0.20$: 0.994 → 0.000	holds
Overall		confirmed

At $T = 0.2$ the dreaming gain persists: ratio 3.58 [3.53, 3.64] at $t = 10$, 3.62 [3.55, 3.68] at $t = 7$. At $T = 0$ the Hebbian estimate falls with size (0.211, 0.179, 0.175, 0.157 for N = 64, 128, 256, 1 000; the thermodynamic value is 0.138) while the dreamed capacity stays ≈ 0.60; the ratio grows for that reason. At loads 0.14 and 0.20 no dream $t = 10$ probe is classified as spurious; at higher loads some are (e.g. all 32 at $\alpha = 0.8$, seed 0), so the claim is load-specific.

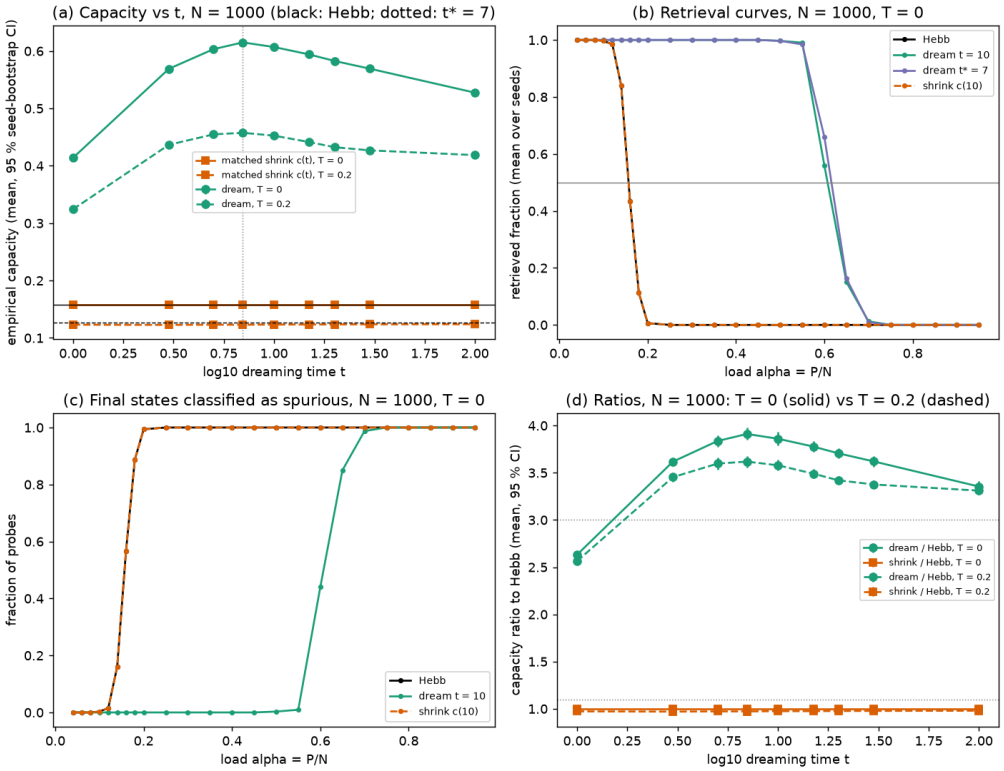


Figure 1. NH-01 confirmatory run, $N = 1\,000$, seeds 0–9 (D2; publication rendering by `code/scripts/publication_figures.py`, the runner's own figure is kept in `data/`). (a) Empirical capacity $\hat{\alpha}_c$ versus dreaming time t for the dreamed kernel (circles) and the matched uniform-shrink control (squares) at $T = 0$ (solid) and $T = 0.2$ (dashed); the horizontal lines are the Hebbian reference; the dotted vertical line marks $t^* = 7$. (b) Retrieved fraction versus load α at $T = 0$ for Hebb, dream $t = 10$, dream $t^* = 7$ and shrink $c(10)$. (c) Fraction of probes whose final state is classified as spurious, versus load, $T = 0$. (d) Capacity ratio to Hebb with seed-level 95 % bootstrap intervals; the dotted lines are the pre-registered thresholds 3.0 (confirmation) and 1.1 (control).

3.2 Correlated patterns

Correlated run (D3), $N = 1\,000$, seeds 100–109, 29 640 cells, $t = 10$ unless stated.

Table 2. $T = 0$, $t = 10$ (C1, C2′).

Level m ($q = m^2$)	Hebb $\hat{\alpha}_c$	dream $\hat{\alpha}_c$	C1 ratio	C2′ t^* (per-seed range); drop to $t = 100$
0 (0)	0.160 [0.158, 0.162]	0.611 [0.606, 0.616]	**3.82 [3.77, 3.88]** — confirmed	7 (5–10); 0.084 [0.079, 0.089] — holds
0.15 (0.0225)	0.068 [0.066, 0.070]	0.616 [0.612, 0.619]	**9.06 [8.84, 9.29]** — confirmed	7 (7–10); 0.088 [0.085, 0.090] — holds
0.30 (0.09)	0.014 [0.014, 0.015] (every seed between loads 0.01 and 0.02)	0.636 [0.633, 0.639]	**44.38 [42.56, 46.97]** — registered C1 confirmed; crossings bracketed by loads 0.01–0.02, magnitude grid-sensitive (see text)	10 (7–10); 0.103 [0.098, 0.108] — holds

Table 3. $T = 0.2$, $t = 10$ (C3′) and level verdict.

Level m	Hebb $\hat{\alpha}_c$	C3′ shrink ratio	Level verdict
0	0.127 [0.125, 0.129]	**0.979 [0.974, 0.984]** — holds	**confirmed**
0.15	0.054 [0.053, 0.056]	**0.986 [0.979, 0.993]** — holds	**confirmed**
0.30	5/10 seeds below grid; zero-coded mean 0.007 [0.003, 0.011]	undefined on 5 seeds; zero-coded five-seed statistic 0.6 [0.2, 1.0], two of its shrink numerators also below grid	registered: confirmed; post-analysis assessment: **inconclusive**

Reading rule for $m = 0.30$. At $T = 0$ all ten Hebbian crossings are bracketed by the loads 0.01 and 0.02 (flags ok; the registration added these two loads for that purpose), so the registered ratio 44.38 is defined and its C1 label is confirmed. Its bootstrap interval describes seed variation conditional on the interpolation rule and does not include the load-grid discretisation error; reading the crossings as anywhere inside their bracketing intervals gives a ratio between about 30 and 65. We therefore do not interpret the magnitude as a precise model-level capacity ratio. At $T = 0.2$ five Hebbian estimates are below the grid; among the five seeds with a positive Hebbian denominator, two shrink estimates are also below the grid. The archived code replaces below-grid capacities by zero, so the Hebbian mean 0.007 [0.003, 0.011] and the five-seed shrink statistic 0.6 [0.2, 1.0]

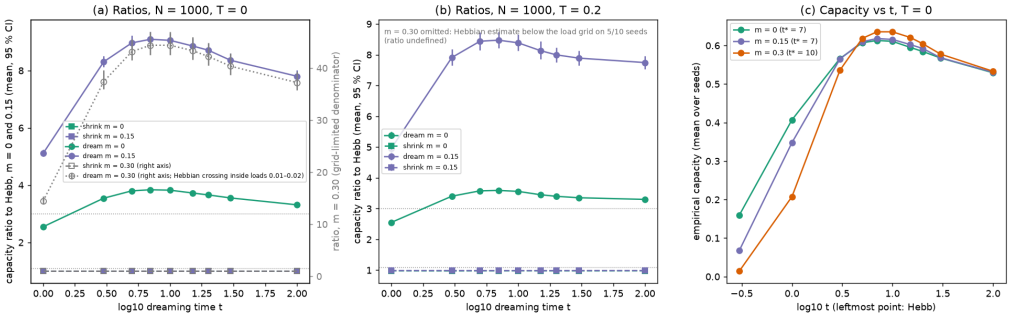


Figure 2. NH-01 correlated-pattern run, $N = 1\,000$, seeds 100–109 (D3; publication rendering by `code/scripts/publication_figures.py`). (a) Capacity ratio to Hebb versus dreaming time at $T = 0$ for ancestor overlap $m = 0$ and 0.15 (left axis; dream: circles, matched shrink: squares, 95 % seed-bootstrap intervals); the $m = 0.30$ curves are drawn in grey on the right axis because their Hebbian denominator is bracketed by the loads 0.01 and 0.02, so the ratio is defined but set by the load grid. (b) Same at $T = 0.2$; $m = 0.30$ is omitted because five of its ten Hebbian estimates, and two further shrink estimates, are below the grid, so its ratios are undefined or zero-coded. (c) Absolute capacity versus t at $T = 0$ for the three levels (leftmost point: Hebb), the robust reading at $m = 0.30$. The unmodified runner figure, whose automatic title labels every level "confirme", is kept in `data/`.

(per-seed 1, 0, 1, 1, 0) are outputs of that coding convention, not censoring-adjusted estimates. The absolute capacities at $T = 0$, also interpolated but less sensitive to the small denominator, are the more robust reading: the dreamed capacity barely moves with correlation (0.611, 0.616, 0.636) while the Hebbian capacity collapses (0.160, 0.068, 0.014), so the paired absolute gain grows 0.451 [0.445, 0.456] → 0.548 [0.543, 0.552] → 0.621 [0.618, 0.625]. The tagged code labels C3′ "holds" and the level "confirmed" in `summary.json`, computed on the five seeds with a positive denominator. This article keeps that registered label and adds, next to it, a post-analysis assessment of inconclusive for the level (decision of 2026-10-30, commit `5884927`): a methodological reservation, not a registered rule. Its ground is censoring: five denominators and two further numerators are below the grid and coded as zero, so the available bootstrap does not estimate the ten-seed control ratio; Hebbian retrieval at $T = 0.2$ survives only within the first load interval. The registered labels are kept unchanged in the raw outputs. The pre-registered below-grid rule was written for C1 only; a future amendment should extend it to C3′ and the load grid below 0.01.

C4 trend: the registered estimate is $R(0.30) - R(0)$ = 40.6 [38.7, 43.2]. The label is `gain_increases`. Its magnitude depends on the interpolation of the small Hebbian denominator at $m = 0.30$ and the interval does not account for load-grid resolution; we retain the increasing direction as a descriptive result only. P4 holds at every level (0 spurious final states under dreaming at $\alpha \in \{0.14, 0.20\}$); the $T = 0$ exactness check passed at every level. $m = 0$ reproduces the independent- pattern run on new seeds (3.82 vs 3.86; $t^* = 7$ in both; shrink 0.979 vs 0.975).

3.3 Internal reproduction of the $N = 1\,000$ studies

Both $N = 1\,000$ runs were re-executed on 2026-10-30 from a fresh clone checked out at the tag, with the SHA-256 of the executed files checked against `git show <tag>:<file>` before launch and the tag record fetched from the GitHub API before launch (D2 and D3, directory `provenance-rerun-20261030/`). Result: **9 120 / 9 120** and **29 640 / 29 640** cells identical to the committed `cells.jsonl.gz` on every field except the wall-clock `task_seconds` (cells matched by key, NaN compared as NaN); `summary.json` identical except its `run_meta` block; `figure.png` byte-identical (SHA-256 `3b64a94e…` and `de3ee720…`). The re-runs used 2 worker processes instead of 4 for the confirmatory

run; the worker count changes only the scheduling.

4 Conclusion

Under decision rules fixed before each study's main run, this partial replication **supports** the central qualitative claims of Fachechi, Agliari and Barra within our finite-size protocol: the dreamed kernel raises the empirical storage capacity of the Hopfield network by a large factor (3.4–3.9 at $t = 10$ for independent patterns, N = 256 to 1 000), leaves 0/320 probes classified as spurious at each of the loads 0.14 and 0.20 in the confirmatory run at $T = 0$, against 51/320 and 318/320 under Hebb (which does not establish the absence of spurious attractors elsewhere in state space), keeps its advantage under thermal noise, and behaves as the projector rule for large t (the pilot's $f = 0$ capacity estimate is right-censored at $\alpha = 0.95$ at the tested times $t \in \{10, 30, 100\}$ for $N \in \{64, 128, 256\}$ — a 50 %-retrieval criterion on at most 32 probed patterns, not a test that every stored pattern is a fixed point). The ratios 3.86 and 9.06 are results of this study, not numbers published in 2019. On correlated patterns the gain grows with correlation at $m \leq 0.15$, consistent with the projector limit's insensitivity to pattern correlations.

Two points go beyond, or qualify, the original. First, the added control shows that the Frobenius-matched uniform rescalings of the Hebbian couplings tested here do not reproduce the gain: they are inert at $T = 0$ by scale invariance and give a mean paired relative capacity reduction of 2.52 % at $T = 0.2$ (seed-bootstrap 95 % interval 1.80–3.35 %). The dreaming effect is a spectral change of the attractor landscape, not a global downscaling at equal synaptic budget. Second, with the self-coupling removed and a noisy cue, the finite-size capacity is not monotone in t: it peaks at $t^* = 7$ and loses 0.088 [0.082, 0.092] by $t = 100$. This does not contradict the mean-field $\alpha_c(t)$: our noisy-cue threshold differs from the equilibrium retrieval boundary of the theory. We observe a lower empirical capacity at $t = 100$ than at the grid optimum $t = 7$ without measuring basin sizes or isolating the cause (self-coupling removal, cue noise, asynchronous dynamics, finite N). The original article does quantify retrieval basins, at $N = 1\,000$ and $\alpha = 0.05$ under parallel zero-noise dynamics, and finds them stabilised by dreaming; that experiment does not cover our higher-load protocol.

Recent work situates these observations. Benedetti et al.[6] follow the eigenvectors of the coupling matrix during unlearning, the spectral reading that distinguishes dreaming from a rescaling in our control. Serricchio et al.[7] combine reinforcement and removal in a "daydreaming" algorithm and report its effectiveness on correlated data generated by a random-features model, a correlation structure different from our single ancestor. Lucibello and Mézard[8] make precise, for dense associative memories, the distinction between the capacity of a typical pattern, of all patterns and the size of the basins — the distinction that also separates our noisy-cue estimator from a test of the stability of every stored pattern.

What this replication cannot settle: absolute capacities (finite-size, estimator-dependent), temperatures other than 0 and 0.2, correlation structures other than a single ancestor, and the $m = 0.30$ level, where the Hebbian crossings lie inside the first load interval (0.01–0.02) at $T = 0$ and below the grid on five seeds at $T = 0.2$. A scale sweep (nerve-wml#90) is open and not included here.

5 Reproducibility

The submission repository https://github.com/hypneum-lab/rescience-nh01 (MIT for the code, CC BY 4.0 for the text and figures) holds the runners (S1), the tests (S2), the three pre-registrations (S3), the three data sets (D1–D3) and the provenance files of the internal re-runs. The runners are the files of `experiments/ nh01_hopfield_dream/`

of hypneum-lab/nerve-wml at the tag `prereg-nh01-correlated-20261027` (commit `df9a1e7`), whose SHA-256 are recorded in `code/CODE_SHA256_AT_TAG.txt`; the pilot tag freezes the pilot protocol only, the code it ran is the first runner commit after it. Software Heritage snapshot `swh:1:snp:01242c8b5873518a503b84a80ce3a302bb69b8a8` (visit 2026-10-29) holds the three tags as releases; the data are archived at Zenodo under DOI 10.5281/zenodo.23833431 (nerve-wml v1.10.0, concept DOI 10.5281/zenodo.19656342).
Environment: Python 3.12.15, numpy 2.5.4, matplotlib 3.11.2 (full freeze in the re-run provenance directory of D2 and D3), single-threaded BLAS (the OpenMP, OpenBLAS, MKL and NumExpr thread counts set to 1), no GPU; host electron-server, 8 logical CPUs (Intel Xeon X5680, no AVX), Linux 6.8.0, shared with other jobs (load 7–20), `nice 19`. Every random draw comes from NumPy's default generator seeded by (20261025, seed, N, stream); bootstrap seed 20261025, 10 000 resamples.
Commands (from `code/`; wall times from each `run_meta.json`):

```
python -m experiments.nh01_hopfield_dream.run \
    --workers 4 --out results/nh01           # D1: 1 547 s
python -m experiments.nh01_hopfield_dream.confirmatory \
    --workers 4 --out results/nh01_confirmatory   # D2: 6 951 s
python -m experiments.nh01_hopfield_dream.correlated \
    --workers 2 --out results/nh01_correlated     # D3: 22 866 s
gzip -k results/<run>/cells.jsonl   # the comparator reads .gz
python -I scripts/compare_nh01.py \
    ../data/<run> results/<run> cmp.json
```

The confirmatory run started 14 s after its tag was pushed (2026-10-25, 13:47:22 UTC); the correlated run on 2026-10-27 at 16:48:06 UTC. The comparator does not set a failure exit status; it writes five fields into `cmp.json`:

```
keys_equal   cells_differing_non_timing   fields_differing
max_abs_diff   summary_equal_excl_run_meta
```

Exact reproduction means matching keys and zero non-timing differences; on another platform, the reported differences are read against the tolerance below. The exit status alone validates nothing. The option `--analyze-only` recomputes `summary.json` and `figure.png` from an existing `cells.jsonl`. Unit tests run in seconds; the pilot takes about 26 min on 8 cores; the two $N = 1\,000$ runs 2–8 h depending on the number of workers (the re-runs with 2 workers took 15 136 s and 30 547 s).
Expected outputs are the committed `summary.json` files. The two $N = 1\,000$ studies were reproduced on the original host with exact non-timing cell values and identical figure files; on another platform, cell keys and discrete outcomes (classifications, flags, sweeps) should match exactly and floating-point diagnostics (`c_fro`, `energy_own`) to about 10^{-12}, while PNG byte identity is not required. The pilot is not covered by the internal re-run.

6 Appendix: provenance table

Id	Content	Location in the submission repository
D1	pilot, $N \in \{64, 128, 256\}$, 28 080 cells	`data/nh01/` (`summary.json`, `cells.jsonl.gz`, `run_meta.json`, `figure.png`, `SHA256SUMS`)

Id	Content	Location in the submission repository
D2	confirmatory, $N = 1\,000$, 9 120 cells, re-run provenance	`data/nh01_confirmatory/` (same files; `provenance-rerun- 20261030/`)
D3	correlated, $N = 1\,000$, 29 640 cells, re-run provenance	`data/nh01_correlated/` (same files; `provenance-rerun- 20261030/`)
S1	runners `run.py`, `confirmatory.py`, `correlated.py` at the correlated tag	`code/experiments/ nh01_hopfield_dream/`; hashes in `code/ CODE_SHA256_AT_TAG.txt`
S2	unit tests (20 functions, 23 cases)	`code/tests/ test_nh01_hopfield_dream.py`
S3	the three pre-registrations, verbatim copies of the tagged files (tag names in `prereg/README`)	`prereg/`
F	publication rendering of the figures	`code/scripts/ publication_figures.py`

7 AI usage

Claude Code (Anthropic) was used as a tool for the code, the documentation and drafts of this text, in sessions driven by the author; every draft was read, corrected and arbitrated by the author. Codex CLI (OpenAI; gpt-6.1-sol, then gpt-6-astra) ran two automated checks of the finished package before submission; the briefs, reports and verdict tables are in `review/`. These checks are not peer review. The design, the three pre-registrations, every verdict and every number in this article were checked and decided by the author. Development history: hypneum-lab/nerve-wml pull requests #81, #86 and #87, issue #68.

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